

Sub-Group:-

1:- which of the following are subspaces of $\mathbb{R}^3$?

Consider, Theorem 4.2.1:

If " W " is a set of one or more vectors in a vector space V , then " W " is a subspace of " V " if and only if the following conditions are satisfied -

- a)- If u and v are vectors in W , then $u+v$ is in W -
- b)- If " k " is a scalar and " u " is a vector in W , then ku is in W -

Consider that :-

$$W = \{(a, 0, 0) : a \in \mathbb{R}\}$$

that is all vectors in $\mathbb{R}^3$ with last two components equal to zero. let

$$U = (u_1, 0, 0)$$

$$V = (v_1, 0, 0)$$

be two vector spaces in $\mathbb{R}^3$. Now adding the two vectors are represented below,

$$U+V = (u_1, 0, 0) + (v_1, 0, 0)$$

$$\Rightarrow U+V = (u_1 + v_1, 0, 0)$$

Therefore, the resulting vector has zeros as the last two components-

For any scalar k , consider the scalar multiple of a vector in " W " as represented below,

$$\begin{aligned} KU &= K(U_1, 0, 0) \\ \Rightarrow KU &= (KU_1, 0, 0) \quad (\text{By Scalar Multiplication}) \end{aligned}$$

(b) Consider that
 $W = \{(a, 1, 1) : a \in \mathbb{R}\}$

that is all vectors in $\mathbb{R}^3$ with last two components equal to 1.

let

$$U = \{U_1, 1, 1\}$$

$$V = \{V_1, 1, 1\}$$

be two vector space in $\mathbb{R}^3$.

Now adding the two vectors as represented below,

$$U + V = (U_1, 1, 1) + (V_1, 1, 1)$$

$$\Rightarrow U+V = (U_1+V_1, 2, 2) \notin W$$

(By vector addition)

Hence, all vectors of the form $(a, 1, 1)$ is not a subspace of $\mathbb{R}^3$ —

(c) :-

$$W = \{ (a, b, c) : a, b, c \in \mathbb{R}, \text{ where } b = a + c \}$$

$$\text{let } U = (a_1, b_1, c_1)$$

$$V = (a_2, b_2, c_2)$$

be two vector spaces in $\mathbb{R}^3$ such that,

$$b_1 = a_1 + c_1$$

$$b_2 = a_2 + c_2$$

$$U+V = (a_1, b_1, c_1) + (a_2, b_2, c_2)$$

$$\Rightarrow U+V = (a_1+a_2, b_1+b_2, c_1+c_2)$$

where,

$$b_1+b_2 = (a_1+c_1) + (a_2+c_2)$$

Hence $U+V \in W$ —

For any scalar K , consider the scalar multiple of a vector in W -

$$KU = K(a_1, b_1, c_1)$$

$$\Rightarrow KU = (Ka_1, Kb_1, Kc_1)$$

where

$$Kb_1 = Ka_1 + Kc_1$$

hence $KU \in W$.

We conclude that all vectors of the form (a, b, c) , where,

$b = a + c$ is a subspace of $\mathbb{R}^3$.

Q2:- which of the following are subspaces of M_{nn} .

a):- The set of all diagonal $n \times n$ matrices.

Let " W " be the set of all diagonal $n \times n$ matrices. Now consider two matrices in W as represented below.

$$A = \begin{bmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{bmatrix}$$

$$B = \begin{bmatrix} b_{11} & 0 & \dots & 0 \\ 0 & b_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & b_{nn} \end{bmatrix}$$

Now finding the sum of the two matrices-

$$A+B = \begin{bmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{bmatrix} + \begin{bmatrix} b_{11} & 0 & \dots & 0 \\ 0 & b_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & b_{nn} \end{bmatrix}$$

$$A+B = \begin{bmatrix} a_{11}+b_{11} & 0 & \dots & 0 \\ 0 & a_{22}+b_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn}+b_{nn} \end{bmatrix}$$

Therefore the result is Hence $A+B \in W$

For any scalar k , consider the scalar

b):- let V be the set of all $n \times n$ matrices A such that $\det(A) = 0$

let " V " be the set of all $n \times n$ matrices A such that $\det(A) = 0$.

let

$$X = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}$$

$$Y = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix}$$

$\Rightarrow X + Y = I$ identity matrix with determinant 1 . So axiom 1 does not hold.

$\Rightarrow V$ is not subspace of M_{nn} .

Subspace.

Q2:-

(c). set of all $n \times n$ matrices A such that $\text{tr}(A) = 0$.

1). let $X, Y \in V$.

$$\begin{aligned}\text{tr}(X+Y) &= \sum_{i=1}^n x_{ii} + y_{ii} \\ &= \sum_{i=1}^n x_{ii} + \sum_{j=1}^n y_{jj} \\ &= \text{tr}(X) + \text{tr}(Y) \\ &= 0 + 0 = 0\end{aligned}$$

$$\Rightarrow X+Y \in V$$

2). let $K \in \mathbb{R}$ be scalar and $X \in V$.

$$\begin{aligned}\text{tr}(KX) &= \sum_{i=1}^n K \cdot x_{ii} \\ &= K \sum_{i=1}^n x_{ii} \\ &= K \cdot \text{tr}(X) \\ &= K \cdot 0 = 0\end{aligned}$$

$$\Rightarrow KX \in V$$

$\Rightarrow "V"$ is a subspace of M_{nn} .

d) :- set of all symmetric $n \times n$ matrices.

1) :- let x and y be matrices in V .

$$\begin{aligned}[x+y]_{ij} &= x_{ij} + y_{ij} \\ &= x_{ji} + y_{ji} \\ &= [x+y]_{ji}\end{aligned}$$

$$\Rightarrow x+y \in V$$

2) - let $k \in \mathbb{R}$ and $x \in V$:

$$\begin{aligned}[kx]_{ij} &= k \cdot x_{ij} \\ &= k \cdot x_{ji} \\ &= [kx]_{ji}\end{aligned}$$

$$\Rightarrow kx \in V$$

$\Rightarrow V$ is a subspace of M_{nn}

e):- set of all $n \times n$ matrices s.t
 $A^T = -A$.

let " V " be the set of all $n \times n$ matrices
 A such that $A^T = -A$.

1):- let $X, Y \in V$:-

$$\begin{aligned}(X+Y)^T &= X^T + Y^T \\ &= X - Y \\ &= -(X+Y)\end{aligned}$$

2):- let $K \in \mathbb{R}$ and $X \in V$:-

$$\begin{aligned}(KX)^T &= K \cdot X^T \\ &= K \cdot (-X) \\ &= -KX\end{aligned}$$

$$\Rightarrow KX \in V$$

$\Rightarrow "V"$ is a subspace of M_n .

f):- The set of all $n \times n$ matrices A for which $Ax=0$ has only the trivial Sol.

let " V " be the set of all $n \times n$ matrices " A " such that $Ax=0$ has only the trivial Sol.

If $Ax=0$ has only the trivial Sol then " A " is invertible matrix; hence $\det(A) \neq 0$

1):- let $X, Y \in V$.

$$X = \begin{bmatrix} 1 & 0 & \vdots & \dots & 0 \\ 0 & 1 & \vdots & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & \vdots & \vdots & \dots & 0 \\ 0 & \vdots & \vdots & \dots & 0 & 1 \end{bmatrix}$$

$$Y = \begin{bmatrix} -1 & 0 & \vdots & \dots & 0 \\ 0 & -1 & \vdots & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & \vdots & \vdots & \dots & 0 & -1 \end{bmatrix}$$

Then $X+Y=O$, since $\det(O)=0$,

$\Rightarrow$ "V" is not a subspace of M_{nn} .
 $O \notin V$

9):- let "V" be a set of all $n \times n$ matrices "A" such that $AB=BA$ for some fixed $n \times n$ matrix B.

1)- let X and Y be matrices from V.

$$\begin{aligned}(X+Y)B &= XB + YB \\ &= BX + BY \\ &= B(X+Y)\end{aligned}$$

$$\Rightarrow X+Y \in V$$

2)- let $K \in \mathbb{R}$ and $X \in V$.

$$\begin{aligned}(KX)B &= K \cdot XB \\ &= K \cdot BX \\ &= B(KX)\end{aligned}$$

$$\Rightarrow (KX) \in V$$

$\Rightarrow$ "V" is a subspace of M_{nn}

Q3:- Subspace of P_3 .

a)- All polynomials $a_0 + a_1x + a_2x^2 + a_3x^3$
for which $a_0 = 0$.

1) - let $f, g \in V$.

$$f(x) = a_3x^3 + a_2x^2 + a_1x$$

$$g(x) = b_3x^3 + b_2x^2 + b_1x$$

$$\begin{aligned}(f+g)(x) &= f(x) + g(x) \\ &= (a_3+b_3)x^3 + (a_2+b_2)x^2 + (a_1+b_1)x\end{aligned}$$

$$\Rightarrow f+g \in V$$

2) - let $K \in \mathbb{R}$ and $f \in V$.

$$f(x) = a_3x^3 + a_2x^2 + a_1x$$

$$\begin{aligned}Kf(x) &= K \cdot (a_3x^3 + a_2x^2 + a_1x) \\ &= Ka_3x^3 + Ka_2x^2 + Ka_1x\end{aligned}$$

$$\Rightarrow Kf \in V$$

$\Rightarrow$ "V" is a subspace of P_3 .

b) :- set of all polynomials of the form $a_0 + a_1x + a_2x^2 + a_3x^3$ for which $a_0 + a_1 + a_2 + a_3 = 0$

1) :- let $f, g \in V$:

$$f(x) = a_3x^3 + a_2x^2 + a_1x + a_0$$

$$g(x) = b_3x^3 + b_2x^2 + b_1x + b_0$$

$$(f+g)(x) = (a_3+b_3)x^3 + (a_2+b_2)x^2 + (a_1+b_1)x + a_0+b_0$$

$$\begin{aligned} & (a_3+b_3) + (a_2+b_2) + (a_1+b_1) + (a_0+b_0) \\ &= (a_3+a_2+a_1+a_0) + (b_3+b_2+b_1+b_0) \\ &= 0+0 = 0 \end{aligned}$$

$$\Rightarrow f+g \in V$$

2) :- let $k \in \mathbb{R}$ and $f \in V$:-

$$f(x) = a_3x^3 + a_2x^2 + a_1x + a_0$$

$$(kf)(x) = ka_3x^3 + ka_2x^2 + ka_1x + ka_0$$

$$\begin{aligned} ka_3 + ka_2 + ka_1 + ka_0 &= k(a_3 + a_2 + a_1 + a_0) \\ &= k \cdot 0 = 0 \end{aligned}$$

$$\Rightarrow Kf \in V$$

$\Rightarrow$ "V" is a subspace of P_3 .

(C):- All polynomials of the form $a_0 + a_1x + a_2x^2 + a_3x^3$ in which a_0, a_1, a_2 and a_3 are rational numbers.

2) let $K = \pi$ and $f(x) = 1 \in V$

$$(Kf)(x) = \pi$$

π is not rational number

$$\Rightarrow Kf \notin V$$

$\Rightarrow$ "V" is not a subspace of P_3 .

d):- All polynomials of the form $a_0 + a_1x$ where a_0 and a_1 are real numbers

1) - let $f, g \in V$,

$$f(x) = a_1x + a_0$$

$$g(x) = b_1x + b_0$$

$$(f+g)(x) = (a_1+b_1)x + a_0+b_0$$

$$a_1, a_0, b_1, b_0 \in \mathbb{R} \Rightarrow (a_1+b_1), (a_0+b_0) \in \mathbb{R}$$

$$\Rightarrow f+g \in V$$

2) - let $k \in \mathbb{R}$ and $f \in V$:

$$f(x) = a_1 x + a_0$$

$$(kf)(x) = k a_1 x + k a_0$$

$$k, a_0, a_1 \in \mathbb{R} \Rightarrow k a_0, k a_1 \in \mathbb{R}$$

$$\Rightarrow kf \in V$$

$\Rightarrow$ "V" is a subspace of P_3 .

Q4:- Subspaces of $F(-\infty, \infty)$?

a):- All functions f in $F(-\infty, \infty)$ for $f(0) = 0$.

1):- let $f, g \in V$,

$$(f+g)(0) = f(0) + g(0) \\ = 0 + 0 = 0$$

$$\Rightarrow f+g \in V$$

a):- let $k \in \mathbb{R}$ and $f \in V$:-

$$(kf)(0) = k \cdot f(0) \\ = k \cdot 0 \\ = 0$$

$$\Rightarrow kf \in V$$

$\Rightarrow V$ is a subspace of $F(-\infty, \infty)$.

b):- $F(-\infty, \infty)$ for $f(0) = 1$

i) let f and g be functions in V ,

$$(f+g)(0) = f(0) + g(0) \\ = 1 + 1 \\ = 2$$

$$\Rightarrow f+g \notin V$$

$\Rightarrow V$ is not a subspace of $F(-\infty, \infty)$

c):- for $f(-x) = f(x)$.

1) let $f, g \in V$.

$$\begin{aligned}(f+g)(-x) &= f(-x) + g(-x) \\ &= f(x) + g(x) \\ &= (f+g)(x) \\ \Rightarrow f+g &\in V.\end{aligned}$$

2):- let $k \in \mathbb{R}$ and $f \in V$.

$$\begin{aligned}kf(-x) &= k \cdot f(-x) \\ &= kf(x) \\ &= (k \cdot f)(x)\end{aligned}$$

$$\Rightarrow kf \in V.$$

$\Rightarrow$ " V " is a subspace of $F(-\infty, \infty)$.

d):- let " V " be the set of all (polynomials of degree 2)

1):- let $f, g \in V$

$$\begin{aligned}f(x) &= x^2 + x + 1 \\ g(x) &= -x^2 + x + 1\end{aligned}$$

$$(f+g)(x) = (1-1)x^2 + (1+1)x + 1+1 \\ = 2x + 2$$

$$\Rightarrow f+g \notin V$$

$\Rightarrow$ "V" is not a subspace of $F(-\infty, \infty)$.

Q5:- which of the following are subspaces of $\mathbb{R}^\infty$?

a):- All sequences "v" in $\mathbb{R}^\infty$ of the form.

$$v = (v, 0, v, 0, v, 0, \dots)$$

1):- let $x, y \in V$:

$$x = (a, 0, a, 0, a, 0, \dots)$$

$$y = (b, 0, b, 0, b, 0, \dots)$$

$$x+y = (a+b, 0, a+b, 0, a+b, 0, \dots)$$

$$\Rightarrow x+y \in V$$

2):- let $k \in \mathbb{R}$ and $x \in V$,

$$x = (a, 0, a, 0, a, 0, \dots)$$

$$Kx = K \cdot (a, 0, a, 0, a, 0, \dots) \\ = (Ka, 0, Ka, 0, Ka, 0, \dots)$$

$$\Rightarrow Kx \in V$$

$\Rightarrow$ "V" is a subspace of $\mathbb{R}^\infty$

b):- $V = (a, 1, a, 1, a, 1, \dots)$

1):- let $x, y \in V$,

$$x = (a, 1, a, 1, a, 1, \dots)$$

$$y = (b, 1, b, 1, b, 1, \dots)$$

$$x + y = (a + b, 2, a + b, 2, a + b, 2, \dots)$$

$$\Rightarrow x + y \notin V$$

$\Rightarrow$ "V" is not a subspace of $\mathbb{R}^\infty$

(c):- $V = (a, 2a, 4a, 8a, \dots)$

1):- let $x, y \in V$

$$x = (a, 2a, 4a, 8a, \dots)$$

$$y = (b, 2b, 4b, 8b, \dots)$$

$$x+y = (a+b, 2a+2b, 4a+4b, 8a+8b, \dots)$$

$$= (a+b, 2(a+b), 4(a+b), 8(a+b), \dots)$$

$$\Rightarrow x+y \in V$$

b):- let $k \in \mathbb{R}$ and $x \in V$ -

$$x = (a, 2a, 4a, 8a, \dots)$$

$$kx = k \cdot (a, 2a, 4a, 8a, \dots)$$

$$= (ka, 2ka, 4ka, 8ka, \dots)$$

$$\Rightarrow kx \in V$$

$\Rightarrow$ "V" is a subspace of $\mathbb{R}^\infty$.

d):- $\mathbb{R}^\infty$, whose components are 0 from some point-

1):- let $x, y \in V$,

$$x = (x_1, x_2, x_3, \dots, x_{n-1}, x_n, 0, 0, \dots)$$

$$y = (y_1, y_2, y_3, \dots, y_{m-1}, y_m, 0, 0, \dots)$$

Without loss of generality we can assume $n < m$

$$x+y = (x_1+y_1, x_2+y_2, \dots, x_n+y_n, y_{n+1}, \dots, y_m, 0, 0, \dots)$$

$$\Rightarrow x+y \in V$$

2) let $k \in \mathbb{R}$ and $x \in V$:

$$x = (x_1, x_2, x_3, \dots, x_n, 0, 0, \dots)$$

$$kx = k \cdot (x_1, x_2, x_3, \dots, x_n, 0, 0, \dots)$$

$$= (kx_1, kx_2, \dots, kx_n, 0, \dots)$$

$$\Rightarrow kx \in V$$

$\Rightarrow "V"$ is a subspace of $\mathbb{R}^\infty$.